GENERAL

1. DESIGN AND DETAILING ARE IN ACCORDANCE WITH BS EN 1992 EUROCODE 2: DESIGN OF CONCRETE STRUCTURES AND BS EN 1993 EUROCODE 3: DESIGN OF STEEL STRUCTURES AS MODIFIED BY THE REQUIREMENT OF THE HONG KONG GOVERNMENT HIGHWAYS DEPARTMENT STRUCTURES DESIGN MANUAL FOR HIGHWAYS AND RAILWAYS 2013 EDITION.

2. DESIGN LOADING IS INACCORDANCE WITH BS EN 1993 EUROCODE 1: ACTIONS ON STRUCTURES AS MODIFIED BY THE REQUIREMENTS OF THE HONG KONG GOVERNMENT HIGHWAYS DEPARTMENT STRUCTURES DESIGN MANUAL FOR HIGHWAYS AND RAILWAY 2013 EDITION.

3. ALL DRAWING ARE TO BE READ IN CONJUCTION WITH THE HONG KONG GOVERNMENT GENERAL SPECIFICATION FOR CIVIL ENGINEERING WORKS 2020 EDITION AND THE PARTICULAR SPECIFICATIONS.

4. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE SPECIFIED.

5. ALL CHAINAGES AND COORDINATES ARE IN METRES. ALL LEVELS ARE IN METRES REFERRING TO THE PRINCIPLE DATUM (mPD)

6. ALL SETTING OUT DIMENSIONS SHALL BE CHECKED PRIOR TO CONSTRUCTION.

7. DO NOT SCALE THE DRAWING.

8. ABBREVIATIONS:

N.T.S.	NOT TO SCALE
S.O.L.	SETTING OUT LINE
CH	CHAINAGE
C/W	CARRIAGEWAY
NB	NOISE BARRIER
C.J.	CONSTRUCTION JOINT
M.J.	MOVEMENT JOINT
G.M.S.	GALVANIZED MILD STEEL
G.I.	GALVANIZED IRON
S.S.	STAINLESS STEEL
SDMHR	SUTRUCTURE DESIGN MANUAL FOR HIGHWAY & RAILWAYS

9. ALL COORDIANTES ARE BASED ON HONG KONG GRID (1980) UNLESS OTHERWISE SPECIFIED.

STRUCTURAL STEELWORK

1. ALL STEELWORKS SHALL BE HOT ROLLED SECTIONS COMPLYING WITH BS 4: PART 1:2005, BS EN 10210-1&2: 2006, BS EN 10034 : 1993, BS EN 10056-1 : 1999, BS EN 10056-2: 1993, BS EN 10067 : 1997 AS APPROPRIATE. STEEL HOLLOW SHALL BE OF GRADE S355J2H. UNLESS OTHERWISE SPECIFIED, OTHER STEEL SECTIONS AND PLATES SHALL BE OF GRADE S355JO IN ACCORDANCE WITH BS EN 10025-2:2004.

2. BEFORE FABRICATION OF STEELWORK, THE CONTRACTOR SHALL SUBMIT A FULL-SET OF WORKSHOP DRAWINGS. TOGETHER WITH DETAILS OF HIS PROPOSED METHOD OF HANDLING. TRANSPORTATION AND ERECTION OF THE COMPLETED STEEL FRAME TO THE SUPERVISOR FOR HIS ACCEPTANCE. ANY MODIFICATION ON STEELWORK TO ACCOMMODATE THE PROPOSED CONSTRUCTION SHALL BE DESIGNED BY THE CONTRACTOR, AND CLEARLY INDICATED IN THE WORKSHOP DRAWINGS FOR THE ACCEPTANCE OF THE SUPERVISOR.

3. AFTER FABRICATION, THE STEELWORK SHALL HAVE ITS SURFACES PROPERLY PREPARED IN ACCORDANCE WITH THE SPECIFICATION REQUIREMENTS AND HOT DIP GALVANIZED TO BS EN ISO 1461:2009 WITH A MINIMUN THICKNESS OF 85 MICRON. ALL RUST, OIL, PAINT AND SURFACE CONTAMINANTS: MILL SCALE AND WELDING SLAG SHALL BE REMOVED BY GRIT BLASTING CUT FACES AND OUTSIDE ARISES SHALL BE GROUND SMOOTH.

4. PROTECTION OF STEELWORKS AGAINST CORROSION SHALL COMPLY WITH EN ISO 12944 AND BS EN ISO 14713.

5. ALL STEELWORK SHALL BE PAINTED IN ACCORDANCE WITH THE COLOUR CODE SEPCIFIED ON RELEVANT DRAWINGS, AND OTHER RELEVANT CLAUSES IN THE SPECIFICAIONS.

6. PAINTING SYSTEM SHALL FOLLOW THE RELEVANT DRAWINGS, AND OTHER RELEVANT CLAUSES IN THE SEPCIFICATIONS.

7. WELDING SHALL BE IN ACCORDANCE WITH BS EN 1011 & 1011-2 AND ELECTODES SHALL BE IN ACCORDANCE WITH BS EN ISO 2560:2009. WELDS SHALL BE SOUND AND CONTINUOUS AVOIDING LOCKED-IN SLAG WELDING.

8. DIMENSION OF FILLET WELDS REFER TO LEG LENGTH OF WELD.

9. WELDING SYMBOLS ARE BASED ON ISO 2553: 1992.

10. WORKMANSHIP SHALL BE IN ACCORDANCE WITH BS 5400: PART 6 INCLUDING AMENDMENTS STATED IN THE PARTICULAR SPECIFICATION FOR STEEL WORK.

11. FASTENERS OTHER THAN STAINLESS STEEL MATERIALS SHALL BE CADMIUM OR ZINC ELECTROPLATED HOT-DIP GALANIZED, OR SHERADIED TO THE APPROPRIATE BRITISH STANDARD, SUBJECT TO THE APPROVAL BY THE SUPERVISOR.

12. NYLON WASHERS OR WASHERS OF OTHER APPROVED INSULATING MATERIALS SHALL BE PROVIDED BETWEEN THE SURFACES OF DIFFERENT METALS, INCLUDING BUT NOT LIMITED TO ALUMINIUM ALLOY, STAINLESS STEEL AND GALVANIZED STEEL.

13. FASTENERS FOR THE STEELWORK SHALL BE ISO METRIC PRECISION HEXAGONAL BOLTS. STUBS AND NUTS SHALL BE ACCORDING TO BS 3692 OR CUP BOLTS TO BS 4933 AS APPROPRIATE, WITH COMPATIBLE WASHERS. EACH FASTENER SHALL BE COMPLETED WITH EITHER A LOCKING NUT OR SPRING WASHER UNLESS SPECIFIED OTHERWISE.

14. UNLESS OTHERWISE SPECIFIED, DIAMETER OF BOLT HOLES SHALL BE AS FOLLOWS: -
    A) 2mm + NOMINAL DIAMETER OF BOLT, FOR BOLTS WITH NOMINAL DIAMETER SMALLER THAN 27mm
    B) 3mm + NOMINAL DIAMETER OF BOLT, FOR BOLTS WITH NOMINAL DIAMETER EQUAL TO OR GREATER THAN 27mm

15. WELDING CONSUMABLE SHALL PRODUCE DEPOSITED WELD METAL HAVING MECHANICAL PROPERTIES NOT INFERIOR TO THE MINIMA OF THE PARENT METALS.

16. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE PROVISION OF ALL PACKING TO ACHIEVE ADEQUATE TOLERANCE AT THE CONNECTIONS.

17. NON-DESTRUCTIVE TESTING OF WELDS SHALL BE APPLIED IN THE FORM OF:
    A) VISUAL INSPECTION TO BS EN ISO 17637 : 2011 FOR ALL WELDS.
    B) SURFACE FLAW DETECTION TO BS EN6072, BS EN 571.
    C) ULTRASONIC EXAMINATION TO BS EN ISO 17640 : 2010 OR RADIOGRAPHIC EXAMINATION TO BS EN 1435.
    D) THE LOCATION AND EXTENT OF TESTING SHALL BE IN ACCORDANCE WITH THE GENERAL SPECIFICATION AND PARTICULAR SPECIFICATION OF CONTRACT

NOISE BARRIER

1. THE CONTRACTOR SHALL SUBMIT SHOP DRAWINGS FOR THE SUPERVISOR'S APPROVAL.

2. BOLTS AND NUTS CONNECTING COMPONENTS OF THE STRUCTURAL FRAME SHALL BE STAINLESS STEEL WITH STRENGTH GRADE A4-80 IN ACCORDANCE WITH BS EN ISO 3506 PART 1 & PART 2. "FORM C" WASHERS SHALL ALSO BE STAINLESS STEEL TO BS EN 10258 AND BS EN 10259 GRADE 316S33 WITH COMPATIBLE NYLON WASHERS.

3. ALL FIXING DETAILS OF NOISE BARRIER PANELS SHOWN ON DRAWING ARE INDICATIVE ONLY. THE CONTRACTOR SHALL DESIGN THE DETAILS AND MATERIALS FOR FIXING THE NOISE BARRIER PANELS TO MAIN SUPPORTING BARRIER POSTS IN ALL NOISE BARRIER TYPES FOR THE SUPERVISOR'S APPROVAL.

4. THE CONTRACTOR SHALL DESIGN ANY NOISE BARRIER SUB-FRAMES NECESSARY FOR THE NOISE BARRIER PANELS AND SUBMIT DESIGN CHECKER CERTIFIED DESIGN CALCULATIONS AND SHOP DRAWINGS TO THE SUPERVISOR'S APPROVAL PRIOR TO FABRICATION.

5. THE DESIGN OF THE NOISE BARRIER SUB-FRAMES AND FIXING SHALL MEET THE FOLLOWING REQUIREMENT:-
    a) THE DESIGN SHALL BE IN ACCORDANCE WITH RELEVANT SECTIONS IN THE LATEST EDITION OF SDMHR AND RELEVANT DOCUMENTS AS REFERRED BY SDMHR.
    B) THE DYNAMIC PRESSURE HEAD SHALL BE 3.3 kpa.

6. NOISE BARRIER PANELS SHALL BE OF PMMA OR POLYCARBONATE MATERIAL COMPLYING WITH ZTV-Lsw 88 OR EQUV.

7. THE PROPOSED NIOSE BARRIER PANELS SHALL INCORPORATE HOLDING DEVICE TO PREVENT PANELS FROM FALLING OFF THE NIOSE BARRIER IN THE EVENT OF ACCIDENTAL IMPACT. THE HOLDING DEVEICE SHALL SATISFY THE REQUIREMENT STATED IN THE HIGHWAYS DEPARTMENT PRACTICE NOTES NO. BSTR/PN/003 REVISION C - NOISE BARRIERS WITH TRANSPARENT PANELS. SUCH HOLDING DEVICE SHALL NOT ADVERSELY AFFECT THE OVERALL APPEARANCE OF THE NOISE BARRIER.

8. COLOUR OF PANELS SHALL MATCH THE SPECIFIED PANTONE COLOUR OR SIMILAR AS CLOSE AS POSSIBLE.THE CONTRACTOR SHALL SUBMIT COLOUR SAMPLES TO THE SUPERVISOR FOR APPROVAL.

9. THE CONTRACTOR SHALL PROPOSE AND DESIGN THE DETAIL AND MATERIALS FOR SEALING THE CAP BETWEEN NOISE BARRIER PANELS AND CONCRETE FOOTING FOR THE SUPERVISOR'S APPROVAL. COLOUR OF SEALING MATERIAL SHALL BE COMPATIBLE WITH THE COLOUR OF THE ADJACENT NOISE BARRIER PANELS.

10. NOISE BARRIER PANEL TYPE AND COLOUR SCHEME REFER TO DRAWING NO. HMWHY2214-BH1001 TO BH1002

11. THE NOISE BARRIER PANELS AND SUB-FRAMES SHALL BE DESIGNED BY THE CONTRACTOR IN ACCORDANCE WITH THE CONTRACT AND THE DESIGN SHALL BE APPROVED BY THE SUPERVISION. FOR THE AVOIDANCE OF DOUBT, NOISE BARRIER PANELS AND SUB-FRAMES SHALL INCLUDE, BUT ARE NOT LIMITED TO SUB-FRAMES, NOISE BARRIER PANELS. FOR NOISE BARRIERS AND ALL THE FIXING DETAILS INCLUDING THE ASSOCIATED FITTINGS, JOINTS, ACOUSTICAL SEALING AND FLASHING DETAILS.

12. ALL PANELS SHALL BE COVERED BY A MANUFACTURER'S GUARANTEE AGAINST WEATHERING AND PERFORMANCE DETERIORATION FOR 10 YEARS MINIMUM.

13. ALL PANELS SHALL BE REPLACEABLE FROM BOTH FRONT AND BACK SIDE.

14. ALL PMMA OR POLYCARBONATE PANELS SHALL BE 20mm THICK.

15. ALL PMMA OR POLYCARBONATE PANELS SHALL BE MOUNTED ON A STRUCTURAL ALUMINIUM FRAME UNLESS SHOWN OTHERWISE. IT SHALL BE DESIGNED AND PROPOSED BY THE CONTRACTOR AND APPROVED BY THE SUPERVISOR PRIOR TO CONSTRUCTION.

16. CLOSING PANEL IS WELL FIXED WITH NO GAP BETWEEN CONCRETE FOOTING, NOISE BARRIER PANELS AND STRUCTURAL STEEL POST. ITS METERIAL ARE GRADE S275 J2H STRUCTURAL STEEL OR SOUND PROOFED AND WEATHER PROOFED MATERIAL AND INSTALLATION DETAILS ARE DESIGNED AND PROPOSED BY THE CONTRACTOR AND APPROVED BY THE SUPERVISOR PRIOR TO CONSTRUCTION.

EXISTING CONDITIONS

1. LEVELS FOR EXISTING GROUND, AT-GRADE CARRIAGEWAYS. CARRIAGEWAYS ON BRIDGES AND OTHER SURFACE FEATURES AS SHOWN ON THE DRAWINGS ARE INDICATIVE ONLY. EXACT LEVELS SHALL BE CHECKED AND VERIFIED BY THE CONTRACTOR, SUBJECT TO THE REVIEW AND AGREEMENT BY THE SUPERVISOR.

2. THE CONTRACTOR SHALL VERIFY AND CONFIRM THE SETTING OUT LOCATIONS OF ALL STRUCTURES AND STREET FURNITURES ON EXISTING GROUND OR EXISTING BRIDGE THAT ARE OF RELEVANCE, INCLUDING THOSE SHOWN AS INDICATIVE ON THE DRAWINGS, AND THOSE NOT SHOWN ON THE DRAWINGS.

TENDER DRAWING

no.	date	description	initial
REVISION			

designed M.T.FUNG SIGNED 31/08/22	drawn C.W.LIU SIGNED 31/08/22
checked K.L.FONG SIGNED 31/08/22	approved C.W.CHAN SIGNED 31/08/22

contract no. HY/2022/14

contract
MODIFICATION WORKS OF NOISE BARRIERS ALONG YUEN LONG HIGHWAY NEAR PARK SIGNATURE

drawing title
GENERAL NOTES

drawing no.	scale
HMWHY2214-BH0001	N.T.S

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MAJOR WORKS
PROJECT MANAGEMENT OFFICE

HIGHWAYS DEPARTMENT HONG KONG